Strategic coexistence theory for evolutionary game theory

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Abstract

Introduction

Understanding the mechanisms that promote the evolutionary stability of competing strategies is a fundamental aim in evolutionary biology. To this end, evolutionary game theory (EGT) and replicator dynamics have provided key foundations, allowing us to predict the outcome of strategic competition through payoff inequalities, invasion criteria, and stability analysis. However, less attention has been given to similarities between replicator dynamics and community ecology until recently (Tarnita and Traulsen, 2025), even though the classic replicator equation stems from the generalized Lotka-Volterra equation.

Interestingly, even a simple two-strategy game can yield heterogeneous outcomes that mirror outcomes in community ecology: the dominance of a singular strategy (prisoner's dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). Such differences in conditions that promote evolutionary stability and dominance of a singular strategy and those that permit strategy coexistence indicate that EGT can be understood from the perspective of coexistence theory. Recasting game dynamics as a coexistence problem immediately raises the following question: when two strategies coexist, does the corresponding mechanism stabilize or equalize competition? Even though EGT allows us to predict whether two strategies can coexist or not, a framework for quantifying coexistence mechanism between competing strategies remains elusive.

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for quantifying coexistence mechanisms. MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to be greater than average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor. In EGT, a strategy's niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions. Thus, stabilizing niche differences arise when each strategy is favored by a different strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each strategy to compete under the limiting strategic environments created by its competitor. This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains. Inspired by this effort, we present a Strategic Coexistence Theory (SCT) for predicting coexistene and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover outcomes from classic, two strategy games, providing reinterpretation of classic results from coexistence-theoretic perspective. We then apply

Strategic coexistence theory

We begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \tag{1}$$

where x_i represents the frequency of strategy i, $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i-th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j. However, because payoffs in many games can be negative, payoff ratios are difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \tag{2}$$

For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i.

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \tag{3}$$

When $1-\rho > 0$, each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other words, intraspecific competition is stronger than interspecific competition.

Second, we define the fitness difference as the ratio between the geometric means of the multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \tag{4}$$

In other words, strategy j has an average competitive advantage over strategy i ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated by itself and its competitor. Then, analogous to MCT, mutual invasion requires the fitness ratio to fall within the bounds set by niche overlap:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \tag{5}$$

This decomposition allows us to understand how different mechanisms contribute to the coexistence or exclusion of competing strategies.

Classic two-strategy games

As an illustrative example, we begin with classic two-strategy games, which can be described by the following payoff matrix (Figure 1A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \tag{6}$$

Here, the first and second rows represent strategies 1 and 2, respectively, and the columns represent the strategy of the interacting opponent. Writing x_1 to denote the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1-x_1), \tag{7}$$
$$\pi_2(\mathbf{x}) = Tx_1 + P(1-x_1). \tag{8}$$

Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1-x_1)\left[(R-T)x_1 + (S-P)(1-x_1)\right]. \tag{9}$$

The signs of $T-R$ and $S-P$ determine whether each strategy can invade the other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition the game space into four regimes (Figure 1B): prisoner's dilemma ($T > R$, $P > S$), hawk–dove game ($T > R$, $S > P$), stag–hunt game ($R > T$, $P > S$), and

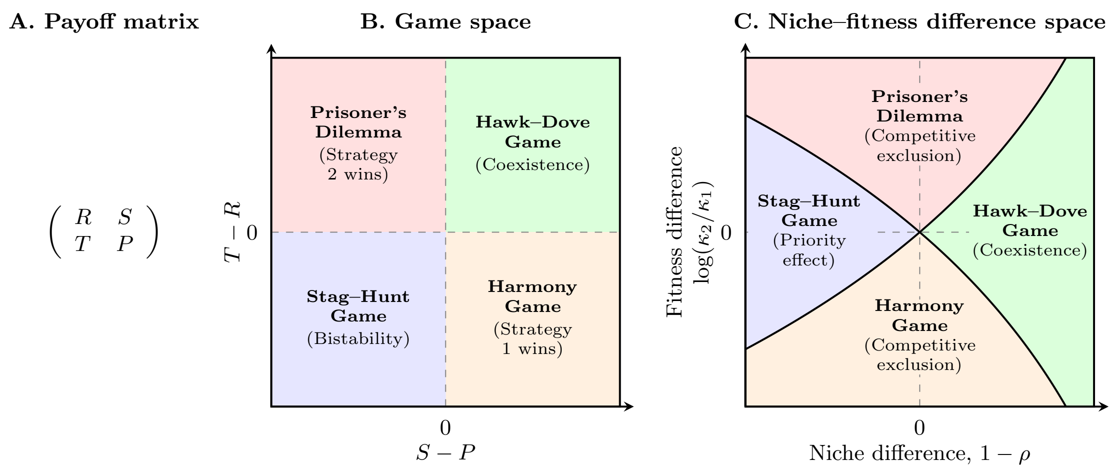

Figure 1: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk–dove games allow coexistence; and stag–hunt games result in bistability. (C) Mapping of the same four games onto niche–fitness difference space. Hawk–dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag–hunt games correspond to the priority-effect regime.

harmony game ($R > T$, $S > P$). In standard EGT, these outcomes are obtained by analyzing the stability of equilibrium conditions. For simple games, this analysis is straightforward.

A coexistence-theoretic approach recovers the same classification but provides a different interpretation (Figure 1C). For a two-strategy game determined by this payoff matrix, the niche difference and fitness difference are given by

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \tag{10}$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \tag{11}$$

As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to $T > R$ and $S > P$. This corresponds to the hawk–dove game, in which each strategy can invade when rare and the two strategies coexist. The remaining games also have natural interpretations in niche–fitness difference space. Prisoner's dilemma and harmony games represent competitive exclusion regimes, where one strategy has a fitness advantage that exceeds niche differences. In contrast, stag–hunt games represent a priority-effect regime, where neither strategy can invade the other and the outcome depends on initial conditions. Therefore, coexistence theory not only allows us to recover the classic results from two-strategy games but also provides new interpretations for these results in terms of niche and fitness differences. Building on this, we next apply this framework to comparing core mechanisms of cooperation.

Comparisons of cooperation mechanisms

Even though cooperative behavior can be found across a wide range of biological systems, a simple prisoner's dilemma shows that cooperation is not an evolutionarily stable strategy. Thus, many researchers have sought to identify mechanisms that promote evolutionary stability of cooperation. Here, we compare five core mechanisms for the evolution of cooperation using SCT and ask how they differ from the perspective of strategy coexistence. Specifically, we consider the following mechanisms: kin selection, direct reciprocity, indirect reciprocity, network reciprocity, and group selection (Nowak, 2006). All of these mechanisms can be described by a simple 2×2 payoff matrix, from which the corresponding niche and fitness differences can be computed using Eq. (10) and Eq. (11).

Building on this, we next apply this framework to microbial cooperation, where nonlinear benefits and environmental context can alter the coexistence of cooperators and defectors.

Microbial cooperation and public goods

Here, we consider a simple, phenomenological model of competition between cooperating and cheating yeast in a cooperative sucrose-metabolism system, where invertase production by cooperators allows cheaters to utilize glucose, a public good released through sucrose hydrolysis (Gore et al., 2009). Specifically, invertase production by cooperators incurs a cost c and generates a total benefit of 1 that is captured privately with efficiency ϵ. Assuming a linear relationship between glucose availability and microbial growth, the payoffs for cooperators π_C and cheaters (defectors) π_D can then be written as

$$\pi_C(f) = \epsilon + f(1-\epsilon) - c, \tag{12}$$
$$\pi_D(f) = f(1-\epsilon), \tag{13}$$

where the use of glucose depends on the fraction of cooperators f. The dynamics of cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1-f)\left[\pi_C(f) - \pi_D(f)\right] \tag{14}$$
$$= f(1-f)(\epsilon - c) \tag{15}$$

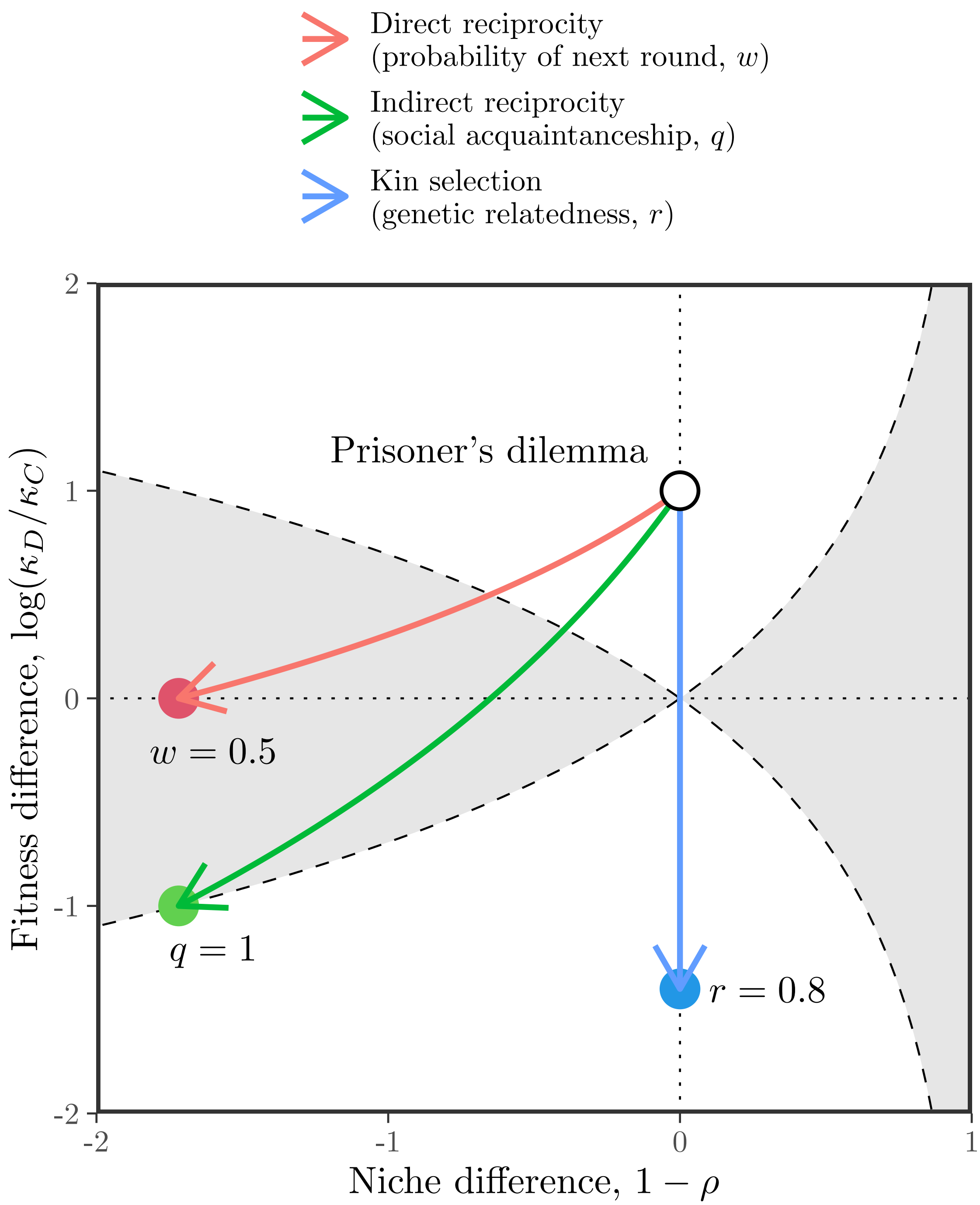

Figure 2: **Comparisons of three mechanisms for the evolution of cooperation using SCT.** ?

For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of f (Figure 2A). Therefore, cooperation is favored when efficiency is greater than the cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than efficiency, $c > \epsilon$ (Gore et al., 2009).

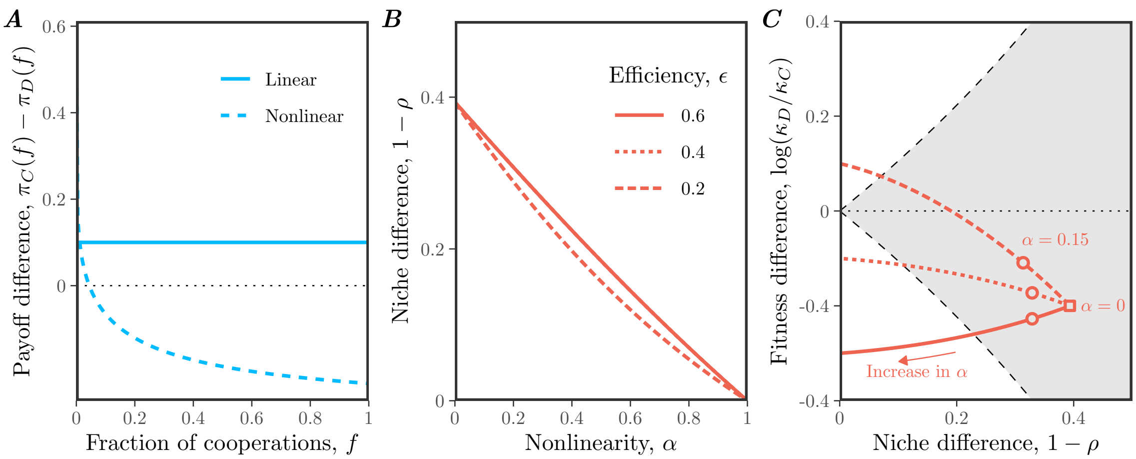

Figure 3: **Caption.** (A) Payoff differences for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.2$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1-\rho$. Each line assumes a different value of efficiency ϵ. (C) Effect of nonlinearity α on both niche difference $1-\rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha = 0$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.2$, while allowing ϵ and α to vary.

Our framework allows us to reinterpret this result from the perspective of coexistence theory. In particular, the two strategies have no niche difference, corresponding to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \tag{16}$$

Therefore, this model always predicts competitive exclusion, except in the neutral case $c = \epsilon$. The outcome is determined entirely by the fitness difference between cooperators and defectors, rather than by stabilizing niche differences.

In real biological systems, the effect of glucose on microbial growth can follow a nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1-\epsilon)]^\alpha - c, \tag{17}$$
$$\pi_D(f) = [f(1-\epsilon)]^\alpha, \tag{18}$$

where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1-f)\left[\{\epsilon + f(1-\epsilon)\}^{\alpha} - c - f(1-\epsilon)^{\alpha}\right]. \tag{19}$$

Under this nonlinear assumption, the payoff difference decreases with the cooperator fraction f, which can promote the coexistence of cooperators and defectors (Figure 2A): cooperators have a competitive advantage when rare, whereas defectors have a competitive advantage when cooperators are abundant.

From a coexistence-theoretic perspective, this means that nonlinear growth can generate sufficient stabilizing niche difference to overcome the fitness difference. In particular, we can show that the niche and fitness differences for this model are given by

$$1 - \rho = 1 - \exp\left(\frac{1-\epsilon^{\alpha} - (1-\epsilon)^{\alpha}}{2}\right), \tag{20}$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1-\epsilon)^{\alpha} - 1 - \epsilon^{\alpha}}{2}\right). \tag{21}$$

These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For example, the cost of cooperation does not affect niche difference. Instead, niche difference is generated by the concave relationship between glucose availability and microbial growth, which increases from 0 to $1 - \exp(-1/2)$ as α goes from 1 to 0 (Figure 2B,C). In contrast, as $\alpha \to 0$, the fitness difference approaches

$$\log(\kappa_D/\kappa_C) \to c - \frac{1}{2}, \tag{22}$$

which no longer depends on the private capture efficiency ϵ (Figure 2C). Thus, strong concavity not only generates stabilizing niche differences, but also acts as an equalizing mechanism (Figure 2C).

Supplementary Text

S1 Derivation of the strategic coexistence theory

To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \tag{S1}$$

where x_i represents the frequency of strategy i, π_i represents the payoff of strategy i, and $\bar{\pi}$ represents the average payoff. Then, the mutual invasion condition for two competing strategies i and j can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \tag{S2}$$
$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \tag{S3}$$

These inequalities will be preserved when we exponentiate the payoffs, $W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x}))$:

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \tag{S4}$$
$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \tag{S5}$$

Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \tag{S6}$$

By multiplying $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$ on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \tag{S7}$$

Thus, by defining niche overlap ρ and fitness difference κ_j/κ_i as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \tag{S8}$$
$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \tag{S9}$$

we obtain the following inequality that defines conditions for mutual invasion and long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \tag{S10}$$

S2 Applications to classic two strategy games

Consider a two strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \tag{S11}$$

There are four classic games that we can define from this simple payoff matrix: prisoner's dilemma ($T > R > P > S$), hawk–dove game ($T > R$, $S > P$), stag–hunt game ($R > T$, $P > S$), and harmony game ($R > T$, $S > P$). Here, we show that each game corresponds to a distinct region in the niche-fitness difference space under SCT.

To do so, we first begin by deriving expressions for niche overlap ρ and fitness difference κ_j/κ_i. Writing x_1 to denote the frequency of strategy 1, the expected payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \tag{S12}$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \tag{S13}$$

Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \tag{S14}$$

$$W_1(0) = \exp(S) \tag{S15}$$

$$W_2(1) = \exp(T) \tag{S16}$$

$$W_2(0) = \exp(P) \tag{S17}$$

Therefore, we have:

$$\rho = \exp\left(\frac{R + P - S - T}{2}\right), \tag{S18}$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \tag{S19}$$

As before, SCT allows us to predict the mutual invasion of two strategy using the following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \tag{S20}$$

Then, simple algebra reveals that each condition in mutual invasion inequality can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \tag{S21}$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \tag{S22}$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \tag{S23}$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \tag{S24}$$

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